

MA587 HW 3

Kyle Hansen

31 March 2026

Question 1 Second order Problem

1(a) First, the differential equation is multiplied by test vector $v \in V$ and integrated over the domain,

$$-\int_0^1 u'' v dx = \int_0^1 f v dx \quad (1)$$

Gauss Divergence theorem is used to reduce the left-hand side,

$$-\left[\cancel{u'v} \Big|_0^1 - \int_0^1 u'v' dx \right] = \int_0^1 f v dx \quad (2)$$

Note: In the original image, there is a handwritten arrow pointing from the text $v(0)=v(1)=0$ to the boundary terms $\cancel{u'v}|_0^1$ in the equation above.

which yields the final weak form,

$$a(u, v) = L(v) \quad \forall v \in V \quad (3)$$

$$a(u, v) = \int_0^1 u'v' dx \quad (4)$$

$$L(v) = \int_0^1 f v dx \quad (5)$$

1(b) To show that a is symmetric, note that the multiplication inside the integral is commutative, so

$$a(u, v) = \int_0^1 u'v' dx = \int_0^1 v'u' dx = a(v, u) \quad (6)$$

and thus a is symmetric. To show that a is continuous, first note that

$$\|u\|_{H_0^1}^2 = \|u\|_{L^2}^2 + \|u'\|_{L^2}^2 \quad (7)$$

$$\|u'\|_{L^2}^2 \leq \|u\|_{H_0^1}^2 \quad (8)$$

$$\|u'\|_{L^2} \leq \|u\|_{H_0^1} \quad (9)$$

and also

$$\|u\|_{L^2} \leq \|u\|_{H_0^1} . \quad (10)$$

By the Cauchy-Schwarz (CS) inequality,

$$\int_0^1 u'v' dx = \langle u', v' \rangle_{L^2} \leq \|u'\|_{L^2} \|v'\|_{L^2} , \quad (11)$$

and thus, using Equation 9,

$$\int_0^1 u'v' dx \leq \|u\|_{H_0^1} \|v\|_{H_0^1} , \quad (12)$$

and a is continuous. To show $L(v)$ is continuous, apply the CS inequality,

$$\int_0^1 f v dx = \langle f, v \rangle_{L^2} \leq \|f\|_{L^2} \|v\|_{L^2} \quad (13)$$

and apply Equation 10,

$$\int_0^1 f v dx \leq \|f\|_{L^2} \|v\|_{H_0^1} . \quad (14)$$

1(c) To show that a is coercive, consider that

$$|v(x)| = \left| \cancel{v(0)} + \int_0^x v'(s) ds \right| \quad (15)$$

$$|v(x)| \leq \int_0^1 |v'(s)| ds = \langle 1, v' \rangle_{L^2} . \quad (16)$$

Then, using the CS inequality,

$$|v(x)| \leq \|1\|_{L^2} \|v'\|_{L^2} \quad (17)$$

$$\|v(x)\|_{L^2} \leq \|(\|v'\|_{L^2})\|_{L^2} \quad (18)$$

$$\|v(x)\|_{L^2} \leq \|v'\|_{L^2} \quad (19)$$

1(d) AI use note: To complete the coding portion of this assignment, I did not use AI to obtain the solution, instead modifying FEniCS' own demo problems. The code for this problem was based on demo_poisson.py. Gemini *was* used to aid in plotting solutions for all three problems.

Some numerical results are displayed in Figure 1, with L_2 and L_∞ error listed in Table 1. Here, the results are within double-precision roundoff error when degree-2 polynomials are used in each cell, since the exact solution is a quadratic polynomial, and thus the finite element method calculates the exact solution, even with only one cell. When using degree-1 polynomials in each cell, the error quickly converges (approximately linearly) when more cells are used.

Table 1: 1D Poisson Results

N_x, P^1	L2 error	Max. pointwise error	N_x, P^2	L2 error	Max. pointwise error
4	5.705e-03	7.812e-03	4	8.037e-17	1.388e-16
8	1.863e-03	2.551e-03	8	1.175e-16	1.804e-16
16	3.566e-04	4.883e-04	16	1.267e-15	1.818e-15
32	8.915e-05	1.221e-04	32	5.075e-15	7.175e-15

Question 2 Poincare Inequality

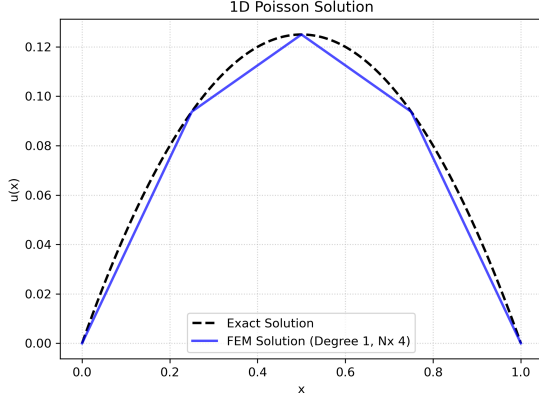
2(a) Similarly to the previous problem, and using the notation $x_1 = x, x_2 =$

$$v(x, y) = \cancel{v(0, y)} + \int_0^x \frac{\partial v(s, y)}{\partial x} ds \quad (20)$$

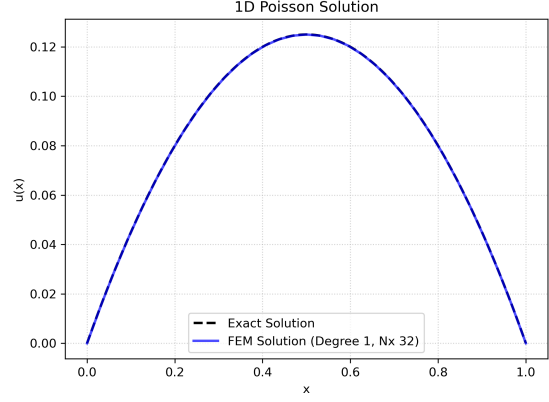
$$v(x, y)^2 = \left(\int_0^x \frac{\partial v(s, y)}{\partial x} ds \right)^2. \quad (21)$$

Then, by the CS inequality,

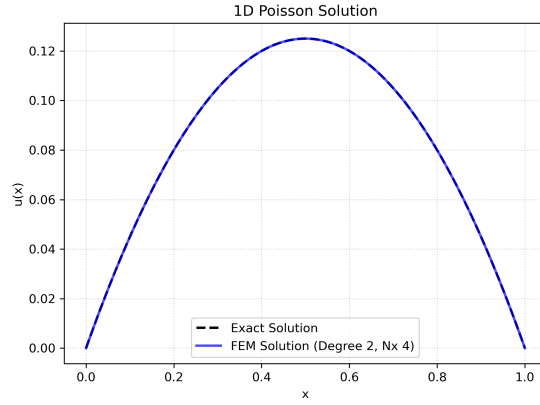
$$v(x, y)^2 \leq \int_0^x \left(\frac{\partial v(s, y)}{\partial x} \right)^2 ds. \quad (22)$$



(a) P^1 , 4 cells



(b) P^1 , 32 cells



(c) P^2 , 4 cells

Figure 1: 1D Poisson Results

Since the term inside the integral is strictly non-negative,

$$v(x, y)^2 \leq \int_0^1 \left(\frac{\partial v(s, y)}{\partial x} \right)^2 ds. \quad (23)$$

Then, since $\|\nabla v\|^2 = \left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2$ and thus $\frac{\partial v^2}{\partial x} \leq \|\nabla v\|^2$,

$$v(x, y)^2 \leq \int_0^1 \|\nabla v\|^2 dx. \quad (24)$$

Integrating this once more over the domain,

$$\int_0^1 v(x, y)^2 \leq \int_0^1 \int_0^1 \|\nabla v\|^2 dx dx \quad (25)$$

$$\int_0^1 v(x, y)^2 \leq \int_0^1 \|\nabla v\|^2 dx. \quad (26)$$

2(b) Note that $\|v\|_{H^2}^2 = \|v\|_{L^2}^2 + \|\nabla v\|_{L^2}^2 + \|\Delta v\|_{L^2}^2$. Begining with the $\|v\|_{L^2}^2$ term, as in the previous part,

$$v(x, y) = \cancel{v_x(\theta, y)} + \int_0^x v_x(s, y) ds \quad (27)$$

$$v(x, y)^2 = \left(\int_0^x v_x(s, y) ds \right)^2. \quad (28)$$

Similarly,

$$v_x(s, y) = \cancel{v_x(x, \theta)} + \int_0^y v_{xy}(s, q) dq, \quad (29)$$

so, plugging this into Equation 28,

$$v(x, y)^2 = \left(\int_0^x \int_0^y v_{xy}(s, q) ds dq \right)^2 \quad (30)$$

Applying the CS inequality twice,

$$v(x, y)^2 \leq \int_0^x \int_0^y v_{xy}(s, q)^2 ds dq. \quad (31)$$

Since the term inside the integral is strictly non-negative,

$$v(x, y)^2 \leq \int_0^\Gamma \int_0^\Gamma v_{xy}(x, y)^2 dx dy. \quad (32)$$

where Γ is the length of the square. Applying Green's formula and using the boundary conditions,

$$v(x, y)^2 \leq \int_\Omega v_{xx} v_{yy} d\mu \leq \int_\Omega |v_{xx}| |v_{yy}| d\mu. \quad (33)$$

Since

$$|\Delta v|^2 = |v_{xx}|^2 + 2|v_{xx}| |v_{yy}| + |v_{yy}|^2, \quad (34)$$

and thus $|v_{xx}| |v_{yy}| \leq \frac{1}{2} |\Delta v|^2$,

$$v(x, y)^2 \leq \frac{1}{2} \int_\Omega |\Delta v|^2 d\mu. \quad (35)$$

Integrating over the domain yields

$$\int_{\Omega} v(x, y)^2 d\mu \leq \frac{1}{2} \Gamma^2 \int_{\Omega} |\Delta v|^2 d\mu, \quad (36)$$

where again Γ is the length of the square. Next, to show the inequality for $\|\nabla v\|_{L^2}^2$, again apply the expansion in Equation 34,

$$|\nabla v|^2 = |v_{xx}|^2 + |v_{yy}|^2 \leq |\Delta v|^2, \quad (37)$$

then integrate over the domain to yield

$$\int_{\Omega} |\nabla v|^2 d\mu \leq \int_{\Omega} |\Delta v|^2 d\mu. \quad (38)$$

Then, combining the inequalities Equation 36 and Equation 38,

$$\|v\|_{H^2} = \|v\|_{L^2} + \|\nabla v\|_{L^2} + \|\Delta v\|_{L^2} \leq (\Gamma^2 + 2) \int_{\Omega} |\Delta v|^2 d\mu. \quad (39)$$

Question 3 Coercivity

3(a) Beginning with the differential form, multiply by $v \in H^1$ and integrate over the domain,

$$-\int_{\Omega} \nabla^2 u \cdot v d\mu = \int_{\Omega} f v d\mu \quad (40)$$

then by the Gauss Divergence Theorem,

$$-\left[-\int_{\Omega} \nabla u \nabla v d\mu + \int_{\Gamma} (\nabla u \cdot \mathbf{n}) v dS \right] = \int_{\Omega} f v d\mu. \quad (41)$$

Using the boundary condition to evaluate the second term,

$$\int_{\Omega} \nabla u \nabla v d\mu - \int_{\Gamma} (g - \gamma u) v dS = \int_{\Omega} f v d\mu. \quad (42)$$

By rearranging terms, the weak form is obtained,

$$a(u, v) = L(v) \quad \forall v \in H^1 \quad (43)$$

$$a(u, v) = \int_{\Omega} \nabla u \nabla v d\mu + \gamma \int_{\Gamma} uv dS \quad (44)$$

$$L(v) = \int_{\Omega} f v d\mu + \int_{\Gamma} g v dS. \quad (45)$$

3(b) To show that a is symmetric, consider multiplicative commutativity (as with problem 1),

$$a(u, v) = \int_{\Omega} \nabla u \nabla v d\mu + \gamma \int_{\Gamma} uv dS = \int_{\Omega} \nabla v \nabla u d\mu + \gamma \int_{\Gamma} vu dS = a(v, u). \quad (46)$$

To show that a is continuous, first note that

$$\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2, \quad (47)$$

then decompose the absolute value using the triangle inequality,

$$|a(u, v)| = \left| \int_{\Omega} \nabla u \nabla v d\mu + \gamma \int_{\Gamma} uv dS \right| \leq \left| \int_{\Omega} \nabla u \nabla v d\mu \right| + |\gamma| \left| \int_{\Gamma} uv dS \right|. \quad (48)$$

For the first part of the integral, take its square and apply the CS inequality,

$$\left| \int_{\Omega} \nabla u \nabla v d\mu \right|^2 \leq \|\nabla u\|_{L^2}^2 \|\nabla v\|_{L^2}^2, \quad (49)$$

and, from Equation 47,

$$\left| \int_{\Omega} \nabla u \nabla v d\mu \right|^2 \leq \|\nabla u\|_{H^1}^2 \|\nabla v\|_{H^1}^2 \quad (50)$$

$$\left| \int_{\Omega} \nabla u \nabla v d\mu \right| \leq \|u\|_{H^1} \|v\|_{H^1}. \quad (51)$$

For the second part of Equation 48, first apply the Trace Theorem,

$$|\gamma| \left| \int_{\Gamma} uv dS \right| \leq C |\gamma| \|uv\|_{L^2(\Omega)} \quad (52)$$

where C is some constant depending on the domain Ω . Then, using the CS inequality,

$$|\gamma| \left| \int_{\Gamma} uv \, dS \right| \leq C|\gamma| \|u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)}. \quad (53)$$

Again using Equation 47,

$$|\gamma| \left| \int_{\Gamma} uv \, dS \right| \leq C|\gamma| \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}. \quad (54)$$

Combining Equation 51 and Equation 54,

$$|a(u, v)| \leq (C|\gamma| + 1) \|u\|_{H^1} \|v\|_{H^1} \quad (55)$$

and thus a is continuous. To show that L is continuous, first apply the CS inequality and Equation 47 to yield

$$\left| \int_{\Omega} fv \, d\mu + \int_{\Gamma} gv \, dS \right| \leq \|f\|_{L^2} \|v\|_{H^1} + \left| \int_{\Gamma} gv \, dS \right|. \quad (56)$$

Then, apply the CS inequality to the last term on the RHS,

$$\left| \int_{\Omega} fv \, d\mu + \int_{\Gamma} gv \, dS \right| \leq \|f\|_{L^2} \|v\|_{H^1} + \|g\|_{L^2(\Gamma)} \|v\|_{L^2(\Gamma)}, \quad (57)$$

then, applying the Trace Theorem,

$$\left| \int_{\Omega} fv \, d\mu + \int_{\Gamma} gv \, dS \right| \leq \|f\|_{L^2} \|v\|_{H^1} + C_0 \|g\|_{L^2(\Gamma)} \|v\|_{H^1} \quad (58)$$

$$\left| \int_{\Omega} fv \, d\mu + \int_{\Gamma} gv \, dS \right| \leq \left(\|f\|_{L^2} + C_0 \|g\|_{L^2(\Gamma)} \right) \|v\|_{H^1} \quad (59)$$

where C_0 is some constant dependent on Ω , and thus $L(v)$ is continuous.

3(c) From the bilinear $a(v, v)$

$$a(v, v) = \int_{\Omega} (\nabla v)^2 \, d\mu + \gamma \int_{\Gamma} v^2 \, dS, \quad (60)$$

$$= \|\nabla v\|_{L^2(\Omega)}^2 + \gamma \|v\|_{L^2(\Gamma)}^2, \quad (61)$$

clearly a is not coercive when $\gamma \leq 0$. Using Friedrichs' Inequality (note that I used Gemini AI to assist with this bound), which states that there is some C_1 for which

$$\|\nabla v\|_{L^2(\Omega)}^2 + \|v\|_{L^2(\Gamma)}^2 \geq C_1 \|v\|_{H^1(\Omega)}^2, \quad (62)$$

the bilinear form is coercive if $\gamma > 1$ (where α is then $= C_1$), though this does not indicate anything about coercivity when $0 < \gamma < 1$.

3(d) The python code for this part was written by modifying the same example problem from dolfinx. Some solutions are displayed in Figure 2, with some errors shown in Table 2, where the number of cells is $2(N_x^2)$. All displayed results were generated with $\gamma = -1.0$.

In general, the solutions with lower γ were less accurate than those with higher γ , as predicted by the analytical results discussed before. With $\gamma < -6$, the more refined meshes had singular systems, and with $\gamma = -10$, even the coarsest mesh did not converge.

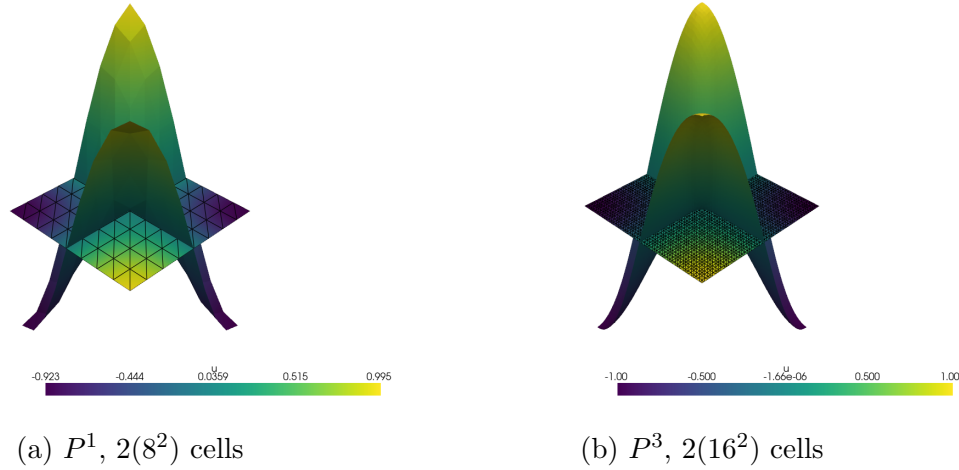


Figure 2: 2D Poisson Results

Table 2: 2D Poisson Results

N_x, P^1	L2 error	Max. pointwise error	N_x, P^3	L2 error	Max. pointwise error
4	1.039e-01	2.538e-01	4	3.189e-04	1.193e-03
8	2.989e-02	8.766e-02	8	1.946e-05	8.442e-05
16	7.796e-03	2.649e-02	16	1.199e-06	5.489e-06
32	1.972e-03	7.555e-03	32	7.449e-08	3.459e-07

The solution converged with all values of

Question 4 Biharmonic Problem

4(a) First, multiply the differential equation by $v \in H_0^2$ and integrate over Ω ,

$$\int_{\Omega} (\nabla^4 u) d\mu = \int_{\Omega} f v d\mu, \quad (63)$$

then apply Gauss Divergence Theorem,

$$\int_{\Gamma} (\nabla^3 u) v dS \xrightarrow{v(\Gamma)=0 \forall v \in H_0^2} - \int_{\Omega} \nabla^3 u \nabla v d\mu = \int_{\Omega} f v d\mu, \quad (64)$$

and applying Gauss Divergence Theorem once more,

$$- \left[\int_{\Gamma} \nabla^2 u (\nabla v \cdot \hat{n}) dS \xrightarrow{\nabla v \cdot \hat{n}|_{\Gamma}=0 \forall v \in H_0^2} - \int_{\Omega} \nabla^2 u \nabla^2 v d\mu \right] = \int_{\Omega} f v d\mu, \quad (65)$$

which yields the weak form of the problem,

$$a(u, v) = L(v) \quad \forall v \in H_0^2(\Omega) \quad (66)$$

$$a(u, v) = \int_{\Omega} \nabla^2 u \nabla^2 v d\mu \quad (67)$$

$$L(v) = \int_{\Omega} f v d\mu. \quad (68)$$

4(b) As in the previous problems, $a(u, v)$ is symmetric by commutativity,

$$a(u, v) = \int_{\Omega} \nabla^2 u \nabla^2 v d\mu = \int_{\Omega} \nabla^2 v \nabla^2 u d\mu = a(v, u). \quad (69)$$

To show that $a(u, v)$ is continuous, first note that

$$\|u\|_{H_0^2}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2 + \|\nabla^2 u\|_{L^2}^2, \quad (70)$$

so

$$\|\nabla^2 u\|_{L^2} \leq \|u\|_{H_0^2}. \quad (71)$$

and

$$\|u\|_{L^2} \leq \|u\|_{H_0^2}. \quad (72)$$

By the CS inequality,

$$a(u, v) = \langle \nabla^2 u, \nabla^2 v \rangle_{L^2} \leq \|\nabla^2 u\|_{L^2} \|\nabla^2 v\|_{L^2} . \quad (73)$$

Using Equation 71,

$$\langle \nabla^2 u, \nabla^2 v \rangle_{L^2} \leq \|u\|_{H_0^2} \|v\|_{H_0^2} \quad (74)$$

and thus $a(u, v)$ is continuous. For $L(z)$, use the CS inequality,

$$L(z) = \langle f, v \rangle_{L^2} \leq \|f\|_{L^2} \|v\|_{L^2} \quad (75)$$

and apply Equation 72,

$$\langle f, v \rangle_{L^2} \leq \|f\|_{L^2} \|v\|_{H_0^2} \quad (76)$$

and thus $L(v)$ is continuous.

4(c) AI usage statement: I provided Gemini with my code from Problem 3, and prompted it with the mixed formulation weak form shown below, then prompted it to modify the Problem 3 code to solve the problem with mixed formulation.

The idea of the mixed formulation is to define some intermediate function w , such the fourth-order PDE becomes a system of two second-order PDEs,

$$-\nabla^2 w = f \quad (77)$$

$$-\nabla^2 u = w \quad (78)$$

$$u = \nabla u \cdot \hat{n} = 0 \quad \text{on } \Gamma. \quad (79)$$

This formulation allows for enforcement of the boundary conditions through the bilinear form.

Using the mixed formualtion, some results are displayed in Figure 3, with some errors shown in Table 3. Here, each increasing the degree of the test and trial functions led to about an order of magnitude of improvement in L_2 error. Additionally, using higher-degree polynomials leads to faster convergence with mesh refinement.

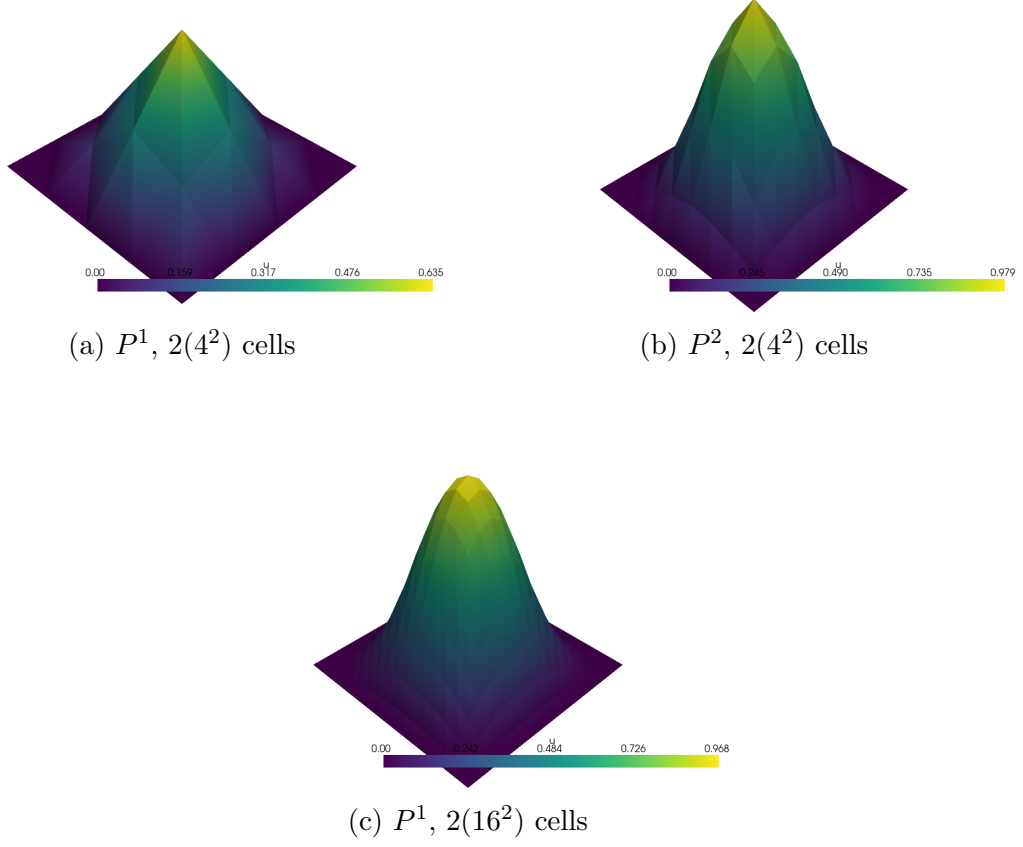


Figure 3: 2D Biharmonic Results

Table 3: 2D Biharmonic Results

N_x, P^1	L2 error	Max. pointwise error	N_x, P^2	L2 error	Max. pointwise error
4	1.701e-01	4.499e-01	4	1.956e-02	6.008e-02
8	5.774e-02	1.728e-01	8	2.044e-03	6.797e-03
16	1.575e-02	4.875e-02	16	2.118e-04	7.854e-04
32	4.029e-03	1.273e-02	32	2.410e-05	8.657e-05
N_x, P^3	L2 error	Max. pointwise error			
4	1.746e-03	6.694e-03			
8	9.373e-05	3.736e-04			
16	5.396e-06	2.203e-05			
32	3.261e-07	1.353e-06			